The Journal of Open Source Software

SpaceAltimetry.jl: A Julia package for working with NASA satellite (laser) altimetry data

Maarten Pronk [1¶] **and Alex Gardner** [2]

1 Deltares, Delft, The Netherlands **2** Jet Propulsion Laboratory, California Institute of Technology, Pasadena, USA ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Adam R. Jensen

Reviewers:

- @ArcticSnow
- @JordiBolibar

Submitted: 13 August 2026
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

Summary

`SpaceAltimetry.jl` is a Julia (Bezanson et al., 2017) package for searching, downloading, and reading spaceborne lidar data from NASA's ICESat, ICESat-2, and GEDI missions. It uses mission-specific granule types and product schemas to expose selected HDF5 variables as lazy [Tables.jl](Quinn et al., 2026)-compatible tables that integrate with the Julia ecosystem.

Statement of need

The lidar satellite altimetry missions by NASA, ICESat (2003-2010) (Schutz et al., 2005), ICESat-2 (2019-) (Markus et al., 2017) and GEDI (2019-) (Dubayah et al., 2020) are of great use to the scientific community as primary sources of high precision surface height and vegetation characteristics, among others (Magruder et al., 2024). The data is distributed as product-specific HDF5 granules, with different tracks, variables, quality flags, and coordinate conventions. Research using these datasets commonly requires searching by product, time, and area of interest; downloading authenticated files; selecting variables and tracks; and writing subsets into local tiling schemes. The Python and R ecosystems have packages for (parts of) these workflows, but the Julia ecosystem does not. `SpaceAltimetry.jl` fills that gap, allowing satellite altimetry data to be easily passed directly into Julia's geospatial and numerical workflows.

The package implements the following functionality:

- **Search**: `search` queries NASA's Common Metadata Repository (CMR) by mission, product, temporal range, and other metadata.
- **Download and sync**: `download` handles authenticated Earthdata downloads for individual granules. For larger jobs, bundled `aria2c` support enables parallel downloads, while `sync` maintains a local folder of granules by scanning existing files and downloading only what is missing.
- **Variable, attribute, and track selection**: Product-specific schemas define default variables, attributes, and tracks for common workflows. Users can add or replace variables with explicit HDF5 paths and transforms, restrict ICESat-2 and GEDI beams at read time, or use a terminal explorer for unfamiliar HDF5 layouts.
- **Lazy tables**: `table(g)` returns an `H5Table` for single-track products (ICESat) or a `PartitionedH5Table` for multi-track products (ICESat-2, GEDI), both implementing the Tables.jl column-access interface. Columns are read lazily from HDF5, while `_FillValue` and `valid_range` metadata become Julia `missing` values, flag variables become `CategoricalVectors`, and scalar attributes are broadcast as `FillArrays`.
- **Filtering and transformations**: Spatial and quality filtering, corrections, and geoid transformations can be chained with Julia's pipe syntax and are specialised to each

DRAFT

product. Each operation declares the columns it needs, allowing `SpaceAltimetry.jl` to read the required HDF5 datasets once before materialisation.

The package is intended for any Julia workflow using NASA satellite laser altimetry data. `SpaceAltimetry.jl` is especially well suited to repeated regional or global processing from local archives, while keeping outputs compatible with Julia's tabular and geospatial interfaces.

State of the field

Several tools already support parts of the spaceborne lidar workflow. SlideRule (Shean et al., 2023) provides a server-side framework for rapid cloud processing of ICESat-2 data, and earthaccess (Barrett et al., 2026) provides a Python package for searching the NASA Earthdata archive.

For local ICESat-2 analysis, icepyx (Scheick et al., 2023) supports querying, obtaining, and manipulating data in Python. The Photon Research and Engineering Analysis Library (PhoREAL) (icesat-2UT, 2026) provides tools and a graphical workflow for ICESat-2 ATL03 and ATL08, while IceSat2R (Mouselimis, 2026) provides ICESat-2 access in R.

For GEDI, gediDB (Besnard et al., 2025) provides a Python toolbox for processing and serving GEDI products. The R package rGEDI (Silva, 2026) supports GEDI visualisation and processing.

`SpaceAltimetry.jl` differs by providing a Julia-native interface and by using the same granule, track-selection, table, and operation concepts across ICESat, ICESat-2, and GEDI. This makes mixed-mission workflows possible without separate mission-specific readers in the user's analysis code, something the Python- and R-based tools above cannot offer Julia users.

Software design

`SpaceAltimetry.jl` is designed around mission abstractions combined with a geospatially aware but generic HDF5-to-table reader. Scientific, mission-specific code remains in product schemas and operations, while the HDF5 table reader is reusable for arbitrary HDF5 files and provides integration with the Julia ecosystem. This design choice was made because early, hard-coded schemas limited extensibility and adoption.

- **Granules** are the central representation of mission files, storing the identifier, URL, local path, metadata, and footprint where available. Granules are parameterised by product, such as `ICESat2_Granule{:ATL08}` or `GEDI_Granule{:GEDI02_A}`, so Julia's multiple dispatch can use them in table construction and filtering.
- **Variables and attributes** describe how HDF5 datasets become table columns. Product schemas provide curated defaults for paths, element types, scalar attributes, and transforms, while users can inspect schemas, append or replace variables, or define schemas for new products with a small number of methods.
- **The H5Table module** is a generic HDF5-to-Tables.jl layer. It resolves dimensions, flattens compatible multidimensional datasets, handles `_FillValue` and `valid_range` metadata, and encodes flag meanings where available.
- **Lazy operations** separate product semantics from HDF5 reading. Filters and transforms declare the variables they need, so missing input columns can be resolved from the product schema before the table is materialised.
- **Julia integration** is provided through interfaces such as `Tables.jl` and `GeoInterface.jl`. Tables can therefore be passed to existing Julia sinks and geospatial tools, with no-data values represented as `missing` and repeated or labelled values represented with `Fill` and `Categorical` vectors where appropriate.

DRAFT

The Journal of Open Source Software

Research impact statement

The package grew out of code developed for the first global lowland digital terrain model derived from ICESat-2 satellite lidar (Vernimmen et al., 2020). `SpaceAltimetry.jl` (then called SpaceLiDAR.jl) was first presented at JuliaCon 2021 (Pronk, M, 2021).

The package was further developed to assess ICESat-2 and GEDI as sources for global terrain modelling (Pronk, Eleveld, et al., 2024). It also underpins DeltaDTM, a global coastal digital terrain model (Pronk, Hooijer, et al., 2024) that used the complete ICESat-2 ATL08 and GEDI L2A archive at the time. It is currently being used for glacier modelling and student research.

AI usage disclosure

AI tools were used during the development of parts of the software and documentation. They helped review implementations, identify bugs or inconsistencies, suggest documentation edits, and improve grammar and spelling, but were not used for the package's overall design, architecture, scientific validation, or interpretation of results.

Acknowledgements

We thank the authors and maintainers of the packages in the JuliaGeo and wider Julia ecosystems on which `SpaceAltimetry.jl` builds.

References

Barrett, A., Battisto, C., Bourbeau, J., Carroll, I., Daniels, C., Fisher, M., Kaufman, D., Kennedy, J., Lopez, L., Lowndes, J., Scheick, J., Steiker, A., & Varghese, S. (2026). *Earthaccess*. https://doi.org/10.5281/zenodo.8365009

Besnard, S., Dombrowski, F., & Holcomb, A. (2025). gediDB: A toolbox for processing and providing Global Ecosystem Dynamics Investigation (GEDI) L2A-B and L4A-C data. *J. Open Source Softw.*, *10*(113), 8593. https://doi.org/10.21105/joss.08593

Bezanson, J., Edelman, A., Karpinski, S., & Shah, V. B. (2017). Julia: A Fresh Approach to Numerical Computing. *SIAM Rev.*, *59*(1), 65–98. https://doi.org/10/f9wkpj

Dubayah, R., Blair, J. B., Goetz, S., Fatoyinbo, L., Hansen, M., Healey, S., Hofton, M., Hurtt, G., Kellner, J., Luthcke, S., Armston, J., Tang, H., Duncanson, L., Hancock, S., Jantz, P., Marselis, S., Patterson, P. L., Qi, W., & Silva, C. (2020). The Global Ecosystem Dynamics Investigation: High-resolution laser ranging of the Earth's forests and topography. *Sci. Remote Sens.*, *1*, 100002. https://doi.org/10/ggjxx8

icesat-2UT. (2026). *Icesat-2UT/PhoREAL*. https://github.com/icesat-2UT/PhoREAL

Magruder, L. A., Farrell, S. L., Neuenschwander, A., Duncanson, L., Csatho, B., Kacimi, S., & Fricker, H. A. (2024). Monitoring Earth's climate variables with satellite laser altimetry. *Nature Reviews Earth & Environment*, *5*(2), 120–136. https://doi.org/10.1038/s43017-023-00508-8

Markus, T., Neumann, T., Martino, A., Abdalati, W., Brunt, K., Csatho, B., Farrell, S., Fricker, H., Gardner, A., Harding, D., Jasinski, M., Kwok, R., Magruder, L., Lubin, D., Luthcke, S., Morison, J., Nelson, R., Neuenschwander, A., Palm, S., … Zwally, J. (2017). The Ice, Cloud, and land Elevation Satellite-2 (ICESat-2): Science requirements, concept, and implementation. *Remote Sens. Environ.*, *190*, 260–273. https://doi.org/10/gg3f7c

Mouselimis, L. (2026). *Mlampros/IceSat2R*. https://github.com/mlampros/IceSat2R

DRAFT

Pronk, M. (2021). *SpaceLiDAR.jl: Processing ICESat-2 & GEDI satellite LiDAR data*. https://pretalx.com/juliacon2021/talk/review/UHKBK8NGULYKURYC778VC8E937YFJHYP

Pronk, M., Eleveld, M., & Ledoux, H. (2024). Assessing Vertical Accuracy and Spatial Coverage of ICESat-2 and GEDI Spaceborne Lidar for Creating Global Terrain Models. *Remote Sens.*, *16*(13), 2259. https://doi.org/10.3390/rs16132259

Pronk, M., Hooijer, A., Eilander, D., Haag, A., de Jong, T., Vousdoukas, M., Vernimmen, R., Ledoux, H., & Eleveld, M. (2024). DeltaDTM: A global coastal digital terrain model. *Sci Data*, *11*(1), 273. https://doi.org/10.1038/s41597-024-03091-9

Quinn, J., Kamiński, B., Anthoff, D., Milan Bouchet-Valat, Papp, T. K., Arakaki, T., abhro, Schouten, R., Robinson, N., mathieu17g, Samuel, O., Carlsson, K., Hoffimann, J., Revels, J., ExpandingMan, Hanson, E., Vogt, C., Anthony Blaom, P., Arslan, A., … Johnson, S. G. (2026). *JuliaData/Tables.jl: V1.13.0*. Zenodo. https://doi.org/10.5281/ZENODO.3376179

Scheick, J., Leong, W. J., Bisson, K., Arendt, A., Bhushan, S., Fair, Z., Hagen, N. R., Henderson, S., Knuth, F., Li, T., Liu, Z., Piunno, R., Ravinder, N., Landung, Sutterley, T., Swinski, J., & Anubhav. (2023). Icepyx: Querying, obtaining, analyzing, and manipulating ICESat-2 datasets. *J. Open Source Softw.*, *8*(84), 4912. https://doi.org/10.21105/joss.04912

Schutz, B. E., Zwally, H. J., Shuman, C. A., Hancock, D., & DiMarzio, J. P. (2005). Overview of the ICESat Mission. *Geophys. Res. Lett.*, *32*(21). https://doi.org/10/bzbspg

Shean, D., Swinski, J. p, Smith, B., Sutterley, T., Henderson, S., Ugarte, C., Lidwa, E., & Neumann, T. (2023). SlideRule: Enabling rapid, scalable, open science for the NASA ICESat-2 mission and beyond. *J. Open Source Softw.*, *8*(81), 4982. https://doi.org/10.21105/joss.04982

Silva, C. A. (2026). *Carlos-alberto-silva/rGEDI*. https://github.com/carlos-alberto-silva/rGEDI

Vernimmen, R., Hooijer, A., & Pronk, M. (2020). New ICESat-2 Satellite LiDAR Data Allow First Global Lowland DTM Suitable for Accurate Coastal Flood Risk Assessment. *Remote Sens.*, *12*(17), 2827. https://doi.org/10/gg9dg6

DRAFT